

Experience Summary:

- 18 years of total experience as C++ (**C++ 0x/11/14**) developer including 2.5 years in Production Support
- Recently worked with Synechron (client - **Macquarie Bank** –COG Tech MTS Market Engineering Group)
- Worked in **Autonomous Vehicle domain (AutoMobility)** and different **Investment banks** more than 8 years including below:
 - ✓ Optel Vision Private Ltd. Goa
 - ✓ LTA (Land Transport Authority) Singapore.
 - Singapore-MIT Alliance for Research and Technology (SMART),Singapore
 - ✓ JP Morgan Chase Bank
 - ✓ UBS Credit Risk IT
 - ✓ Chicago Bank of Exchange (CBOE)
 - ✓ BNP Paribas
 - ✓ Credit Suisse
- Worked with renowned companies included Accenture Singapore, Cognizant (CTS Singapore & India), Headstrong Services Pvt .Ltd. India, CMC Ltd. India

Technical Skill:

- **Programming :** C++ (**C++ 0x/11/14**), VC++, **Python**, Unix, Shell Scripting,
- **Advance Concepts:** **STL, Multithreading, Functor, Lambda Expression** & other C++11/14 advanced concepts.
- **Unit testing:** Follow **TDD** with **Gmock/GTest**.
- **Version Control Systems:** GIT, SVN
- **Tools:** Clion, GDB, DBX, Toad, JIRA

Professional Qualification:

Bachelor of Technology (**IT**) from Uttar Pradesh Technical University (**UPTU**) Lucknow, UP, India (year 2000-2004).

Personal Traits

- Good communication, Organizational and Management Skills
- Able to balance and prioritize multiple tasks.
- Confident and Consistent in producing results.
- Hard Working, Self-Confident, Solution oriented approach.
- Self-analyzing & logic driven.
- Ability to manage and to work with groups

Freelancer Assignment:

C++ SDK Developer & Technical Writer @ Agora Inc | Jan 2023 – Sep 2024

Tools: C++ 11/14, GIT, JIRA, UNIX, Windows, VSCode, MDX

Project Description:

Agora is a pioneer and global leader in Real-Time Engagement, providing developers with simple, flexible, and powerful APIs, to embed real-time voice, video, interactive streaming, chat, and artificial intelligence capabilities into their applications.

As a technical writer and C++ SDK Developer at Agora.io, I am responsible for creating and reviewing documentation for all Agora products including messaging, voice, and video solutions. I contribute to several short-term and long-term technical writing projects, covering topics such as APIs, SDKs, cloud services, video conferencing, and live streaming.

Key Contributions:

- Developed and documented SDKs for video calling, voice calling, chat, and signaling across platforms (**Linux, Windows, Unreal Engine, Web, Unity, iOS, macOS**).
- Authored quick-start guides, API references, and deployment documentation to simplify integration for developers.
- Wrote demo code for SDK implementation on Windows/Linux (C++) and Unreal Engine.
- Designed and debugged Linux/C++ SDKs, conducted unit testing, and optimized performance.
- Collaborated with cross-functional teams to ensure seamless integration and robust system design.

Relevant Project Experience (Most Recent First)

Macquarie Bank –COG TECH CGM MTS Market Operation Engineering Group @ Synechron, Gurgaon

Tools: C++ 11/14, Python, GIT, JIRA, UNIX

Duration: April 2023 to Feb2024

Project Description:

Corporate Operations Group (COG) drives the market operational effectiveness of Macquarie and supports its global business.

Role & Responsibilities:

I was owning a complete module “Future Matchout” responsible for matching of trades of Global Clearing System (GCS) and Macquarie Trading System (MTS). The main purpose of the Matchout Process is to essentially replicate in Hoops what the Macquarie Global Clearing System(GCS) has already done when matching trades based on settlement confirmation from its designated clearers,

A Matchout occurs when new trade dealt have been closed out in full or partially against existing short or long position in the same contract. The Matchout process will create deliveries against open deals that will result in Cash flows in Hoops after Create real is run, which the accountants will use to reconcile to the general ledger (GL) . The GL entries have already been generated by the Matchout process performed by GCS.

I need to work on related new features, Bug fixing & other customer requirements. Apart from that during my tenure, I have drafted the understanding doc by re-engineering of complete module for different region Trade system (Business group).

Work for C++ based Pharma product @Optel Group, Goa

Tools: C++ 11/14, Visual Studio, GIT, JIRA, Jenkins, GMock/GTest

Duration: Sep 2020 to May 2022

Project Description:

OPTEL is a leading global provider of traceability, track-and-trace and vision systems for a growing number of industries, mainly pharmaceuticals. It provide real time supply chain visibility through it's innovative track & trace solutions. It offer true end to end traceability, providing granular data of every step of supply chain starting from raw material to the consumer and beyond. Through serialization & digitization, Optel connect all the players in the supply chain.

Role & Responsibilities:

I have worked in 3 **products** managed by windows based desktop applications developed in C++. I need to work on related new features, Bug fixing & other customer requirements. My day to day Responsibilities included:

- Worked on assigned task (stories/Bug fix/Task) for a epic with other team members. Also owned some epic single handily.
- As a mandatory step, drafted proper documents for each step of epic progress, starting from **Functional Requirement** and **Approach/Design documents**.
- Before start of actual work I need to share these documents to the product owner, architecture & other developers, to get there review to finalize the design.
- Follow **TDD (Test Driven Development)** approach to draft **proper unit test** for my each work of development using **gmock/gtest**.
- Developer side testing of local release before raising the PR (peer review).
- To create Alpha release for SQA testing.
- Update release document for my changes (fix/new feature) & put the functionality & design in confluence for the reference of other stakeholders..
- To do **PR(peer review)** and participate on **design discussion** for another developer's Epic/story.
- Scrum participation & to attend day to day meetings with Product owner, other developers, Agile facilitator, clients/customers and other stakeholders.

Remarkable work:

Logger Management from scratch for one product.

The "Crash Report" Library:

A 'generic' C++ library which provides a way for Windows applications to register for handling run time errors and generate a password protected crash report zip file when a crash occurs. It contains a crash dump, log files and audit/production trail databases, system information XML file etc. The crash report will be automatically generated and the application should be terminated, without any manual intervention from the user. Due to the wealth of the information contained in the crash report, it is envisaged that the crash report will simplify the troubleshooting process in the event of an unforeseen crash/difficult to reproduce crash occurring at the customer site. This is directly in line with our goal of reducing operation costs and saving time. This would help reduce the time to respond and potentially help with the resolution time as well. There is less back and forth needed and there is minimal need to access the production line.

Singapore Train Ticketing Machine @Land Transport Authority (LTA Singapore)

Duration: September 2019 to July, 2020.

Tools: C++ 11/14, Python, GDB, Clion, GIT, Ubuntu

Project Description:

The Land Transport Authority (LTA) spearheads land transport developments in Singapore. It is responsible to plan, design, build and maintain Singapore's land transport infrastructure and systems.

Currently working on Core back-end C++ programming for SG Train Ticketing Machine.

Role & Responsibilities:

- Single hand working on end to end (From Interface to Core Backend development including designing, testing, bug fixing) for new feature 'CEPAS Token Concession Pass'. It covers all aspect starting from GUI Design to handling back-end server's communication (query and update) through Web Socket APIs and applying the core logic to handle application in C++. Data flow and thread handling are the most crucial part of application.
- Documented the design and functionality on a regular basis along with class diagrams & EA diagrams.
- Prepared and documented the test cases and test scenarios for unit testing against all the requirements.
- Adept in end-to-end development of various applications from requirement analysis to system study, designing, coding, testing, de-bugging, implementation and documentation.

Agent based Traffic Simulator (Auto Mobility) (Driver less Vehicles) @ SMART Singapore

Duration: May 2017 to July 2019

Tools: C++ 11/14, Python, GDB, Clion, PGAdmin, GIT, Python,

Project Description: SimMobility is a traffic simulation platform prototype for Autonomous Future Mobility. Aim of the project to work with Singapore Government Transport Authority to implement Autonomous Vehicles (**Driver less Vehicles**) in Singapore and Boston as well.

Role & Responsibilities:

- Worked as core C++ Developer on **Autonomous Vehicles (Driver less Vehicles)** Module.
- Worked on On-Call/On-Hail -controllers, Break Module, Parking Module and other related Algorithms & implementation.
- Worked on day to day Change-Requirements/Enhancements/Issues/Bug Fixing/Data Base, Memory & Performance Issues.
- Worked on many reports & statics generation on every frame tick of different agents like, driver, pedestrians & passengers.
- Updated Containers & logics to get good performance with new features of C++11 & STL.
- Implemented many advanced C++ features in legacy code to get good performance.

CONCORDE (C++) to ATHENA (python) Migration @JPMorgan Chase @Cognizant, Singapore

Duration: Feb 2016 to April 2017

Tools: C++, Python UNIX, Shell Scripting, Sybase, Web eReporting tool, SVN, Toad, JIRA.

Role & Responsibilities:----- C++ / Python Individual Contributor

- Working for Concorde to Athena Migration of downstream feeds & eRepoting.
- Involved in all daily Production Support Activities and L2/L3 support for Middle Office & Front

Trader Office.

- Responsibility included being part of discussion with different stake holders of different related business to understand requirement for Migration and other required enhancements.
- Worked for Automation of different Trade book activities/processes.
- Worked to maintain system process SOD/EOD/Reval/Daily Daemon

UBS Credit Risk IT@ Accenture, Singapore

Duration: June 2014 to Dec-2015

Tools: C++ 11/14 /Solaris-Linux/VC++/Oracle. SVN, Toad, JIRA, SNOW/Shell Scripts/ Autosys

Role & Responsibilities ----- C++ Developer as well as L3 Support.

- Worked as C++ Developer for Calculator Server which is responsible for all calculation of the system, like Yield, Yield Curve, Time Point, EvaluationPoint, Dirty Price, Positions, etc.
- Involved in Technical/Functional documentation
- Involved in analysis for any job failure or system crash in environment Production /Preprod /Dynamic /F2bi).
- Responsible to debug and fix the issues found in Database, shell scripts, C++ code or any Autosys job definition.
- In parallel I had worked in Solaris to Linux porting of current application. Existing Project was compatible for windows and Solaris. I had to put changes to make it portable for Linux (RHEL6). Making changes in Makefiles, scripts (build, inc, xml and others), used macros, libraries and shared objects and related changes in code. Recompile the existing ACE/TAO, PHI, NAG and other supporting packages.

BNP Paribas bank @ Accenture, India

Duration: Nov 2012 to June 2014

Tools/Environment: Solaris/Linux/TNG/Autosys,/Oracle/Shell scripts/ Service Now/Toad/Jira

Role & Responsibilities ----- Production Support as well as Unix/Shell Script Developer

- Involved in development of automation scripts.
- Provide fix for coming defect/issues related to ongoing jobs and DB operations.
- Understanding of complete flow of different jobs/ processes to provide on spot solution and permanent fix.

ALIP (Accenture Life Insurance Platform) @ Accenture, India

Duration: April, 2012 to Nov, 2012

Tools: VC++/C++ 11 /Core Java UNIX /DBX, Makefile, SVN, Toad, Shell Scripting

Role & Responsibilities ----- C++ Developer

- Extensively used STL & Multithreading.
- Worked mainly on Debugging & fixing for system crash. Core dump Analysis(DBX)
- Involved in Coding, Bug fixing and testing the module.
- Technical/Functional documentation including SRS, design, delivery & testing documents.

Telefonica @ Accenture, India

Duration: Oct, 2011 to April 2012
Tools: C++, Java UNIX /DBX/Makefile, Shell Scripting

Role & Responsibilities ----- C++Developer

- Worked as Individual contributor as well as Team Lead.
- Involved in the entire Bugs/Issues related R&D.
- Technical/Functional documentation

Concord BAU @JPMorgan Chase @ Cognizant, India

Duration: July, 2010 to Sep, 2011
Tools: C/C++, Core Java SYBASE, Windows/Linux/Solaris, DBX, Shell Scripting, Makefile, SVN, AveiSQL.

Role and Responsibilities -----C++ Developer as well as Application Support.

- Involved in Coding, Bug fixing, Enhancement, Investigation and testing the module.
- Also involved with JNI Debugging with JAVA client.

CBOE-Linux Porting Phase-I @ Chicago Board Options Exchange @Cognizant (CTS).

Duration: March, 2010 to July, 2010.
Tools: C++/ Core Java Linux, Solaris, GDB, Makefile, Shell Scripting.

Role and Responsibilities ----- C++/Solaris-Linux Porting

- Standalone Single handed ported the complete code.
- Involved in Coding, Bug fixing and testing the module.
- Involved in all the R&D work related to get the proper result on execution of pStats utility.
- Worked in Control M Validation tool development in JAVA.

Tree-Tester Automation Tool @ Credit Suisse @ Cognizant (CTS).

Duration: November, 2009 to March, 2010.
Tools: C++, VC++ 2006/08. VSS, STL, Advance Excel Programming.

Role and Responsibilities -----C++ Developer.

- Involved in R&D to perform the Jobs, Coding for XML Generator Utility, which generate XML.
- Involved in all phases of Module Development, Client Interaction, Designing, Coding, and Debugging along with Unit/Integration Testing.

RC\Migrator @ DI Corporation, New Delhi.

Duration: May, 2008 – October 2009.
Tools: VC++, Oracle, IBM Purify

Role and Responsibilities ----- C++ Developer

- a. Involvement in DDL- Generation module and Object Space module.
- b. Involvement in Bug fixing of different module, Developing & Designing DDL-Parser Module.
- c. Involved in all the phases of the Module Development, Client Interaction, Designing, Coding, and Unit/Integration Testing.

Client Connectivity @Head Strong, Noida.

Duration: August, 2007 to May, 2008.

Tools: Unix/Shell Scripting/JIRA/Service-Now/Oracle

Role and Responsibilities ----- Production Support

- Worked on Production Support team resolving to handle Issues Prod /Dev/ Live/UAT environments
- Involved in fixing the bugs for day to day system jobs & system crash (Database and Shell scripts related.)

I-Policy Security Manager

❏ I-Policy Network @ iPolicy Networks Limited

Duration: October, 2006 – July, 2007.

Tools: C++-Solaris/VC++/Oracle/IBM Purify/ BUGZILA, Shell Scripting

Role and Responsibilities ----- C++ Developer as well as Support

- Added new Alarms/Alerts for existing modules (FW, IDS, URL) and for new module (AV) using the specified message format. Socket communication was used over TCP/IP via for relaying the alarm/alert.
- Involved on resolving Issue reported by L1 team on High/Critical/Blocker and Showstopper Issues

Integrated Air Command Control System @IAF @ CMC Limited

Duration: Dec 2005 to October, 2006.

Tools: C, Solaris, GDB, Make, GTK, POSIX

Role and Responsibilities ----- C Developer

- Involved in coding using Message Queues/Socket/Multithreading in C.
- Involved in client interaction for requirement gathering & prepared related Documents of SRS, design & testing)

Online Travel Pricing @Electrobug Technologies @CMC Limited

Duration: July, 2004 to December, 2005.

Tools: VC++, Oracle, EBW Tool.

Role and Responsibilities ----- C++ Junior Programmer

- Mainly involve in testing the module developed by Developer
- Involved in different stages of developing and testing the module under senior developer.
- Involved in developing the small module related to URL Generation using C++ and Oracle

Personal Information:

Father's Name: Mr. Mahesh Shankar Gupta

Language Known: English, Hindi

***Thanks
Pankaj Gupta***